

How to Get into **FPGA**

The FPGA business is growing at a rapid pace and so is the technology. A bachelor's degree in electronics, electrical or computer science and, of course, a good understanding of the topic is all it takes to launch your career in FPGA



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The job situation in field-programmable gate arrays (FPGAs) has seen an uptrend in the last few years due to increase in their use in different areas. An FPGA is an integrated circuit designed to be configured by the customer or designer after manufacture. The FPGA configuration is generally specified using a hardware description language (HDL).

FPGAs were first introduced into the market about 25 years ago and back then, it was a very simple technology. In the last few years, there are two main factors that have changed the landscape for the better.

"Earlier, people could easily create any logic on FPGAs, but now these are more technical in nature," says Sanjay Churiwala, director, ADVS, Xilinx India.

"FPGAs today come with high-speed interfaces and serial communication protocols. Since these are more capable technically, they allow users to develop complex products and designs that were not possible before. Application-specific integrated circuits are an alternative to FPGAs that were commonly used before. But not many markets can expect such high returns and hence more people opt to work on FPGAs instead. As a result, there has been a huge increase in job opportunities for FPGA engineers,"

Churiwala adds.

Today, design development at the block level is mostly independent of target implementation. Vasudevan Aghoramoorthy, vice president, semiconductor and systems, Wipro Technologies, explains, "FPGAs have increasingly become the choice for new technology and low-volume products across markets like telecommunication infrastructure, industrial automation, aerospace and defense, and next-generation consumer electronics."

As per semiconductor industry data, the total global programmable market was worth close to \$5.1 billion in 2011, with \$4 billion coming from FPGAs.

"We expect India to grow as a market for engineers equipped with FPGA skillset. In the coming years, engineers equipped with FPGA skillset will be in demand at two levels—technology/block development and implementation. From a technology standpoint, such skillset would be needed for block development and validation of custom differentiating and reusable hardware logic. Secondly, from an implementation standpoint, they will be in demand to achieve optimisation for high device utilisation and performance," Aghoramoorthy adds.

The popular FPGAs available in the market are manufactured by companies like Xilinx, Lattice and Altera. Apart from FPGA manufacturers, candidates can also consider working for organisations that design around FPGAs. These include Cisco, Juniper, Samsung and Tejas Networks. Firms like National Instruments, Wipro and